

SIMATS ENGINEERING

ASSESSMENT TOOL 1

COURSE CODE: CSA51

COURSE NAME: Cryptography and Network Security

COURSE OUTCOME COVERED

CO2: Analyze Public Key crypto system strategies using RSA and key Exchange for Encryption algorithm to strengthen information security. (BL4,BL5)

Analytical Problem Solving: 40%

QUESTIONS:

Total Marks: 50

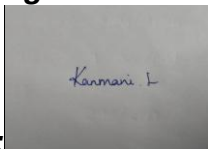
Annexure A

Code of Conduct:

I Kanmani L/192511414 (Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment.

I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the student



Annexure A

Analytical Problem Solving

1. RSA Key Generation and Security Analysis

Given:

$p = 17, q = 19$

Compute n and $\phi(n)$

Choose $e = 5$ and calculate the private key d

Encrypt the message $M = 7$

Decrypt the ciphertext and verify correctness

Analyze the impact of choosing a small prime number on RSA security

2. ElGamal Encryption Analysis

Given:

$p = 23, g = 5, \text{ private key } x = 7, \text{ random } k = 3, \text{ message } M = 10$

Compute the public key

Perform encryption to generate ciphertext pair $(C1, C2)$

Decrypt the ciphertext to retrieve the original message

Analyze how randomness (k) improves security in ElGamal

3. RSA Digital Signature Verification

Given:

$p = 11, q = 17, e = 7, \text{ message } M = 8$

Compute n and $\phi(n)$

Find private key d

Generate the digital signature for the message

Verify the signature using the public key

Analyze how digital signatures ensure non-repudiation

4. Diffie-Hellman with Attack Resistance Evaluation

Given:

$p = 29, g = 2$

User A private key = 5

User B private key = 12

Compute public keys for both users
Compute the shared secret key
Analyze what happens if an attacker intercepts and replaces public keys
Suggest a cryptographic solution to ensure authenticity

5. Hybrid Encryption System Analysis

Given:

A system uses RSA for key exchange and AES for data encryption

RSA encryption time = 5 ms per key exchange

AES encryption time = 0.5 ms per data block

Total data blocks = 500

Calculate total time taken for encryption

Compare with using only RSA for encrypting all data

Analyze why hybrid encryption is more efficient

Justify its use in modern secure communication systems

RUBRICS FOR CALCULATION:

Criteria	Weightage	Level 4 (Exemplary)	Level 3 (Proficient)	Level 2 (Developing)	Level 1 (Beginning)
Problem Understanding	20%	Clearly interprets problem, identifies all parameters and requirements accurately	Understands problem with minor gaps	Partial understanding; misses key elements	Misinterprets or unable to understand problem
Method Selection	20%	Selects most appropriate method/formula with strong justification	Selects correct method with minor errors	Inappropriate or partially correct method	Unable to select method
Solution Process	25%	Logical, systematic steps with correct approach throughout	Mostly correct steps with minor mistakes	Disorganized steps; multiple errors	No clear process
Accuracy of Calculation	20%	All calculations accurate; correct final answer	Minor calculation errors	Frequent errors; incorrect final answer	No valid calculations
Interpretation of Results	15%	Clearly interprets results with units and engineering relevance	Basic interpretation with minor omissions	Limited or unclear interpretation	No interpretation

RSA Key Generation and Security Analysis

Given:

- Prime numbers: $p=17, q=19$
- Public exponent: $e=5$
- Message: $M=7$

Step 1: Compute n and Euler's Totient Function

RSA uses two prime numbers to generate the modulus.

$$n=p \times q = 17 \times 19 = 323$$

Euler's Totient Function:

$$\phi(n) = (p-1)(q-1) = (17-1)(19-1) = 16 \times 18 = 288$$

Therefore,

- Modulus (n) = 323
- Euler's Totient = 288

Step 2: Calculate the Private Key (d)

The private key satisfies

$$e \times d \equiv 1 \pmod{\phi(n)}$$

Substituting values,

$$5d \equiv 1 \pmod{288}$$

The multiplicative inverse of 5 modulo 288 is

$$d=173$$

because

$$5 \times 173 = 865 = 3(288) + 1$$

Hence,

- Public Key = (5,323)
- Private Key = (173,323)

Step 3: Encryption

Encryption formula:

$$C = M \bmod n$$

Substituting,

$$C = 75 \bmod 323 \quad 75 = 16807 \quad 16807 \bmod 323 = 317$$

Hence,

$$\text{Ciphertext} = 317$$

Step 4: Decryption

Decryption formula:

$$M = C^d \bmod n \quad M = 317^{173} \bmod 323$$

Using modular exponentiation,

$$M = 7$$

Thus, the decrypted message is the same as the original message.

Security Analysis

RSA security depends on the difficulty of factoring the modulus n .

When small prime numbers such as 17 and 19 are used:

- The modulus (323) can be factored instantly.
- An attacker can determine p and q .
- The private key can be easily calculated.
- Such keys are completely insecure.

Modern RSA implementations use 2048-bit or 3072-bit keys, making factorization computationally infeasible.

Conclusion

RSA successfully encrypts and decrypts the message. However, the use of small prime numbers makes the system vulnerable to factorization attacks. Therefore, large prime numbers are essential for secure RSA implementation.

2. ElGamal Encryption Analysis

Given

- Prime number $p=23$
- Generator $g=5$
- Private key $x=7$
- Random number $k=3$
- Message $M=10$

Step 1: Compute the Public Key

The public key is

$$y = gx \bmod p \quad y = 5^7 \bmod 23 \quad y = 17$$

Hence,

Public Key = (23,5,17)

Step 2: Encryption

First,

$$C_1 = g^k \bmod p \quad C_1 = 5^3 \bmod 23 = 10$$

Next,

$$y_k = 17^3 \bmod 23 = 14$$

Then,

$$C_2 = M \times y_k \bmod p \quad C_2 = 10 \times 14 = 140 \bmod 23 = 2$$

Therefore,

$$\text{Ciphertext} = (10, 2)$$

Step 3: Decryption

Compute

$$S = C_1 \times \text{mod } p \quad S = 107 \bmod 23 = 14$$

The inverse of 14 modulo 23 is 5.

Therefore,

$$M = C_2 \times 14^{-1} \bmod 23 = 2 \times 5 = 10$$

Thus,

Recovered Message = 10

Security Analysis

ElGamal uses a random value k during every encryption.

Advantages:

- Produces different ciphertexts for the same plaintext.
- Prevents frequency analysis.
- Enhances semantic security.
- Protects against replay attacks.

If the same random value k is reused, attackers can recover the private key or decrypt messages.

Conclusion

ElGamal correctly encrypts and decrypts the message. The use of a unique random number k for every encryption is the main reason for its strong security.

3. RSA Digital Signature Verification (10 Marks)

Given

- $p=11, q=17$
- $e=7$
- Message $M=8$

Step 1: Compute n and $\phi(n)$

$$n=11 \times 17=187 \quad \phi(n)=10 \times 16=160$$

Step 2: Compute the Private Key

Find

$$7d \equiv 1 \pmod{160}$$

The inverse is

$$d=23$$

Therefore,

Public Key:

$$(7, 187)$$

Private Key:

$$(23, 187)$$

Step 3: Generate Digital Signature

The signature is

$$S = M^d \pmod{n} \quad S = 8^{23} \pmod{187} = 173$$

Hence,

$$\text{Digital Signature} = 173$$

Step 4: Verification

Verification uses

$$M = S \cdot \text{mod}^{-1} \cdot 1737 \cdot \text{mod} 187 = 8$$

The recovered message equals the original message.

Hence,

Signature is valid.

Analysis

RSA digital signatures provide:

- Authentication: Verifies the sender's identity.
- Integrity: Detects message modifications.
- Non-repudiation: The sender cannot deny signing the message.
- Security: Only the private key owner can generate the signature.

Conclusion

RSA digital signatures ensure secure authentication, data integrity, and non-repudiation, making them widely used in digital certificates, software signing, and secure communications.

4. Diffie-Hellman with Attack Resistance Evaluation (10 Marks)

Given

- $p=29$
- $g=2$
- User A private key = 5
- User B private key = 12

Step 1: Compute Public Keys

For User A,

$$A = 2^5 \cdot \text{mod} 29 = 3$$

For User B,

$$B = 212 \bmod 29 = 7$$

Public keys:

- User A = 3
- User B = 7

Step 2: Compute Shared Secret

User A computes

$$K = 7^5 \bmod 29 = 16$$

User B computes

$$K = 3^{12} \bmod 29 = 16$$

Thus,

Shared Secret = 16

Attack Analysis

Diffie-Hellman does not authenticate exchanged public keys.

If an attacker intercepts and replaces public keys:

- User A establishes a secret with the attacker.
- User B establishes another secret with the attacker.
- The attacker can decrypt, modify, and re-encrypt messages.

This is known as the Man-in-the-Middle (MITM) attack.

Solution

Authentication methods include:

- Digital signatures
- Public Key Infrastructure (PKI)
- Digital certificates

- Authenticated Diffie-Hellman
- TLS/SSL certificates

These techniques verify the identity of communicating users and prevent MITM attacks.

Conclusion

Diffie-Hellman securely generates a shared key but must be combined with authentication mechanisms to resist active attacks.

5. Hybrid Encryption System Analysis

Given

- RSA key exchange time = 5 ms
- AES encryption time = 0.5 ms per block
- Total blocks = 500

Step 1: Hybrid Encryption Time

RSA is used only once for exchanging the AES key.

RSA time

=5 ms

AES time

$500 \times 0.5 = 250$ ms

Total time

$5 + 250 = 255$ ms

Hence,

Hybrid Encryption Time = 255 ms

Step 2: RSA Only

If RSA encrypts every block,

$$500 \times 5 = 2500 \text{ ms}$$

Therefore,

RSA Only Time = 2500 ms

Comparison

Method	Time
Hybrid Encryption	255 ms
RSA Only	2500 ms

Hybrid encryption is nearly 10 times faster.

Analysis

Hybrid encryption combines the strengths of asymmetric and symmetric cryptography:

- RSA securely exchanges the encryption key.
- AES encrypts the actual data efficiently.
- RSA is computationally expensive and unsuitable for large data.
- AES is optimized for fast, bulk encryption while maintaining strong security.

Applications

Hybrid encryption is used in:

- HTTPS (TLS/SSL)
- VPNs
- SSH
- Secure email (PGP)
- Cloud storage
- Online banking and e-commerce

Conclusion

Hybrid encryption offers both high security and excellent performance. By using RSA only for secure key exchange and AES for encrypting large amounts of data, it significantly reduces encryption time while maintaining confidentiality. Hence, it is the preferred approach in modern secure communication systems.